

A1-4RD | V1 electronics shopping list

Rebuilt for what you already own or can access: soldering supplies, heat shrink, 1 k-ohm resistors, LEDs, jumper wires, a microSD reader, a multimeter, a breadboard, and white or black PLA for internal prints. This is the linked, priced shopping list for one Pi Zero 2 W prototype with two moving servos, I2S audio, a button, and a status LED. US list prices checked 24 Sep 2026; click part names for product pages.

Group	Buy	Qty	Cost
Compute	Pi Zero 2 W, soldered header	1	\$20.75
Compute	2x20 female Pi socket	1	\$1.50
Compute	32 GB microSD card	1	\$19.95
Audio	SPH0645 I2S mic breakout	1	\$6.95
Audio	MAX98357A I2S amp breakout	1	\$5.95
Audio	8-ohm, 1 W mini speaker	1	\$1.95
Motion	SG92R micro servo	3	\$17.85
Controls	16 mm momentary pushbutton	2	\$1.90
Power	USB-C 5 V sink breakout, CC resistors	1	\$2.95
Power	5.1 V, 3 A USB-C power supply	1	\$8.80
Power	2200 uF, 16 V bulk capacitor*	1	\$3.32
Build	Half-size solderable Perma-Proto	1	\$4.50
Build	40-pin breakaway male header	1	\$1.95
Build	22 AWG stranded ribbon, 10 wires	1 m	\$4.25
Build	26 AWG stranded ribbon, 10 wires	1 m	\$2.50
Build	8 A, 2-pin screw terminal	2	\$2.20
Build	M2.5 nylon standoff/screw kit	1 kit	\$14.95
Bench	Pi micro-USB power supply	1	\$8.80
Finish	Panchroma Metallic Bronze PLA, 1.75 mm	1 kg	\$24.99
Order-now subtotal			\$156.01

Cost and scope

Buy now: \$156.01 before shipping and tax, including one bronze PLA spool for visible prints. Three servos means two installed and one spare; two buttons means one installed and one spare. Packs leave extras. Allow another **\$20-\$35** for the final input/servo protection, small decoupling parts, enclosure fasteners, and cable strain relief after measurements and CAD. That puts the purchased electronics and filament at roughly **\$176-\$191** before shipping/tax, or about **\$200-\$240 at checkout** across five vendors. These are planning estimates, not quotes.

Already owned, so excluded: soldering supplies, heat shrink, 1 k-ohm resistors, LEDs, jumper wires, microSD reader, multimeter, breadboard, and white or black PLA for internal prints. No custom PCB is ordered for this first build; the solderable protoboard carries the initial circuit.

Selection gates

Audio: the SPH0645 mic and MAX98357A amp are test candidates. Verify simultaneous capture and playback on the Pi's shared I2S controller before committing to the enclosure or a PCB.

Power: the 5.1 V / 3 A supply is a test candidate for the assembled unit. Measure Pi voltage while both servos start/reverse and audio plays. The listed USB-C sink breakout has the required 5.1 k-ohm CC pull-down resistors; the earlier breakout did not.

***Bulk capacitor:** buy it for power experiments, but do not put 2200 uF directly on USB-C VBUS. Decide its placement and the attach/inrush protection after measuring the load.

Mechanical: confirm SG92R torque and the jaw/neck envelope. The small speaker is a fit candidate; limit amplifier output to its 1 W rating. Final M3 enclosure hardware depends on the printed design.

Filament: Panchroma Metallic Bronze is a bronze-colored PLA, not a metal-filled filament. Use it for visible parts and accessible white/black PLA for internal parts. Confirm your printer takes 1.75 mm filament and has a 0.4 mm or larger nozzle.

Test equipment

Use a current-limited 5 V bench supply and an oscilloscope for current and voltage-transient measurements; borrow lab equipment if available. Neither instrument is included in the electronics checkout estimate. The listed Pi micro-USB supply powers the Pi alone during bring-up.